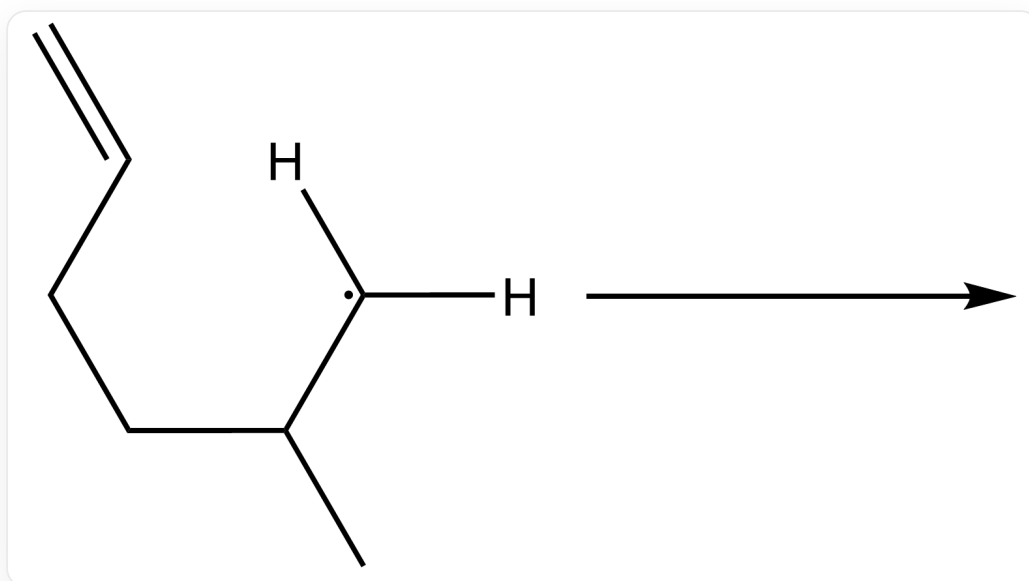
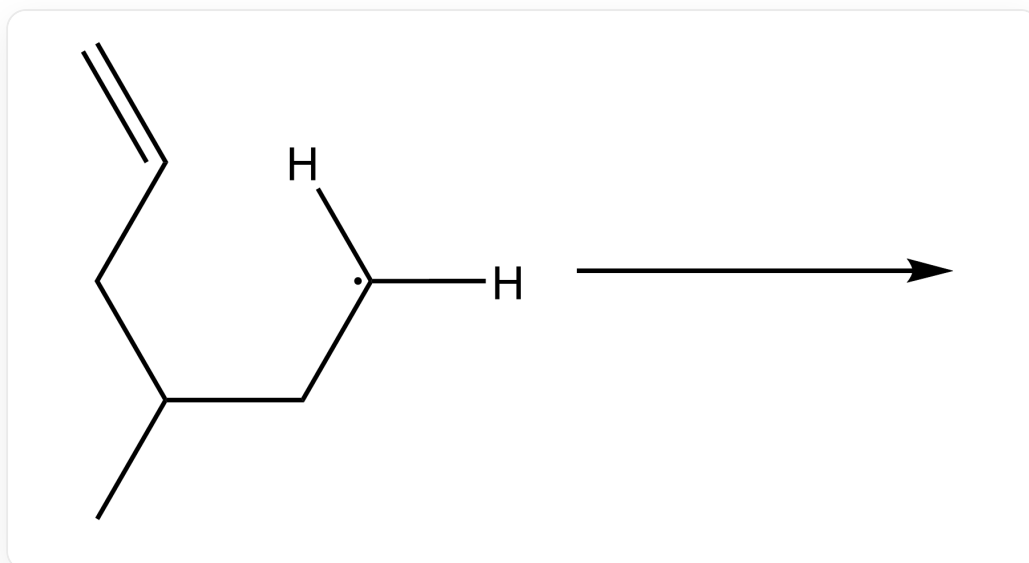


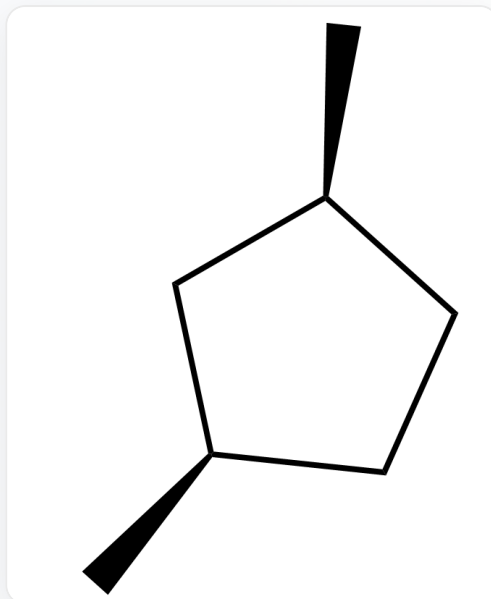
Question

The radical cyclization process sometimes exhibits certain stereoselectivity:



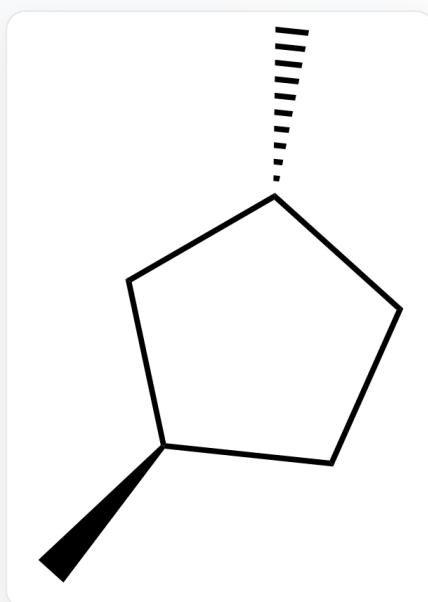
Given that the system contains Bu_3SnH , please predict the product structures of the above two reactions respectively and determine whether they have enantiomers.

A.



C[C@@H]1C[C@H](C)CC1

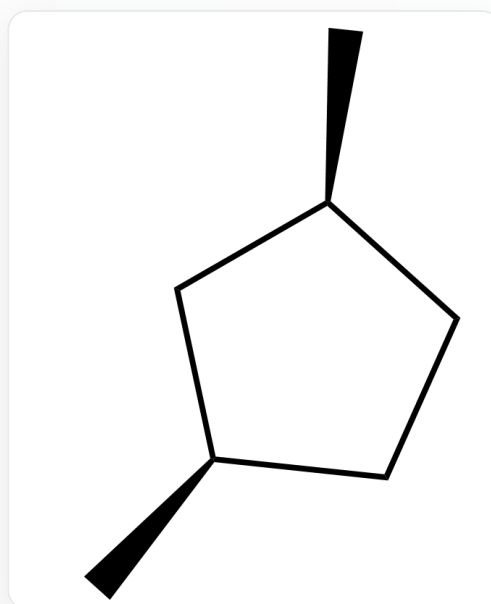
No enantiomers



C[C@H]1C[C@H](C)CC1

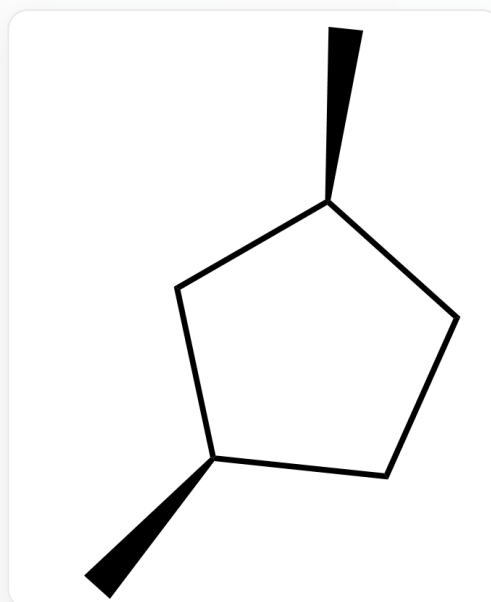
Has enantiomers

B.



C[C@@H]1C[C@H](C)CC1

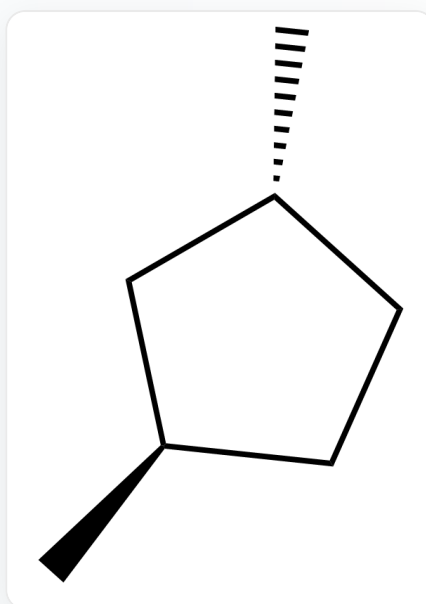
No enantiomers



C[C@@H]1C[C@H](C)CC1

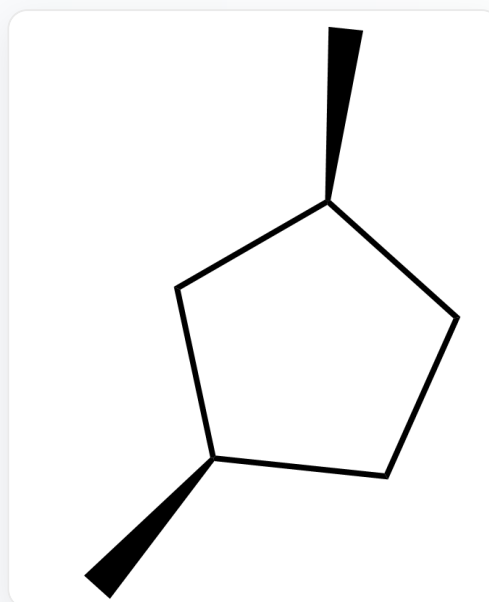
No enantiomers

C.



C[C@H]1C[C@H](C)CC1

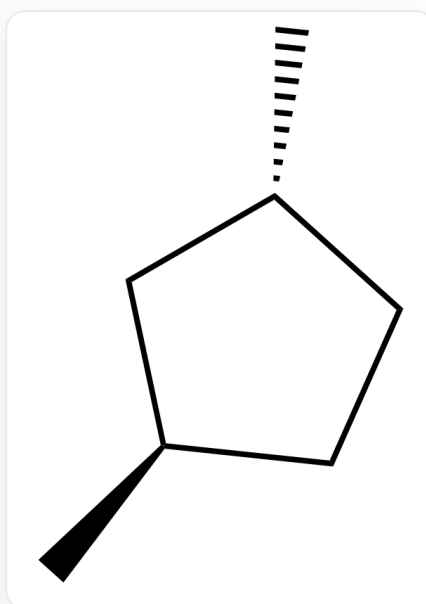
No enantiomers



C[C@@H]1C[C@H](C)CC1

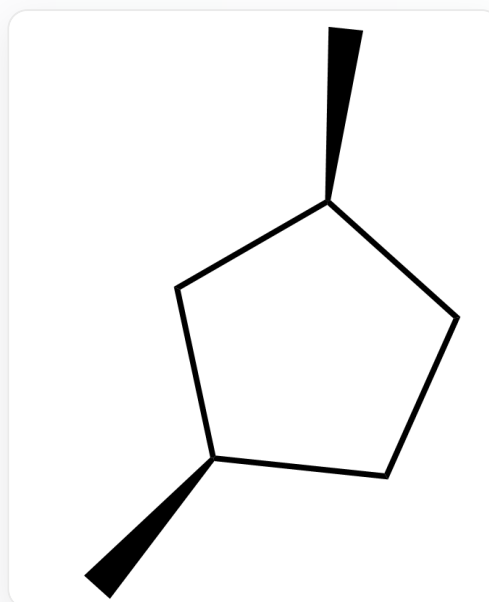
No enantiomers

D.



C[C@H]1C[C@H](C)CC1

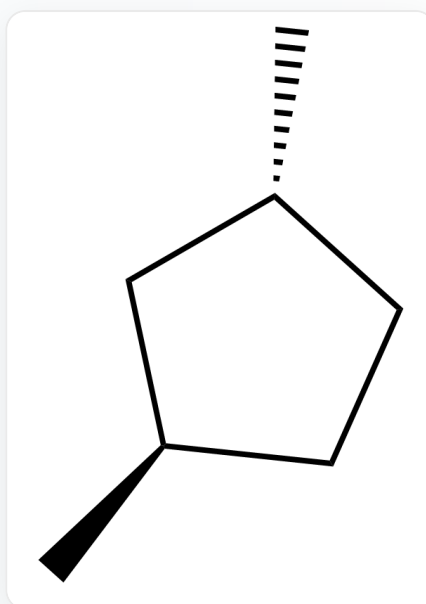
Has enantiomers



C[C@@H]1C[C@H](C)CC1

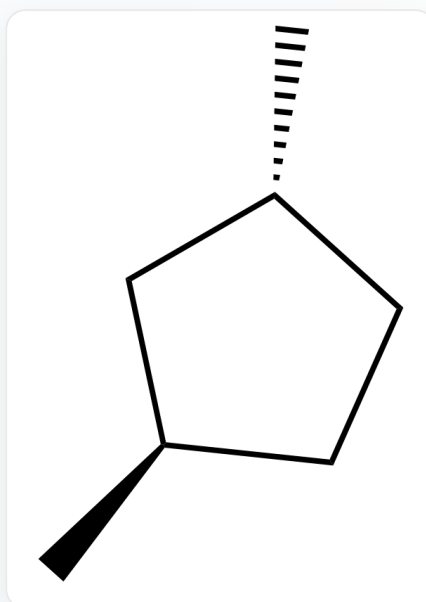
No enantiomers

E.



C[C@H]1C[C@H](C)CC1

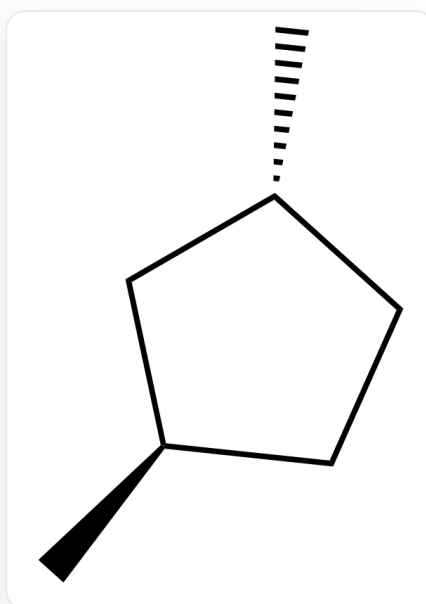
No enantiomers



C[C@H]1C[C@H](C)CC1

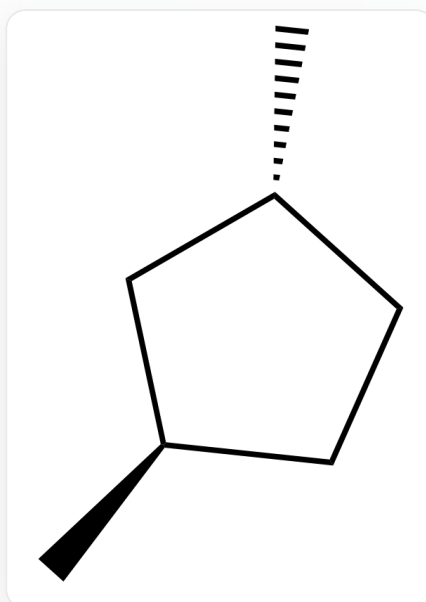
No enantiomers

F.



C[C@H]1C[C@H](C)CC1

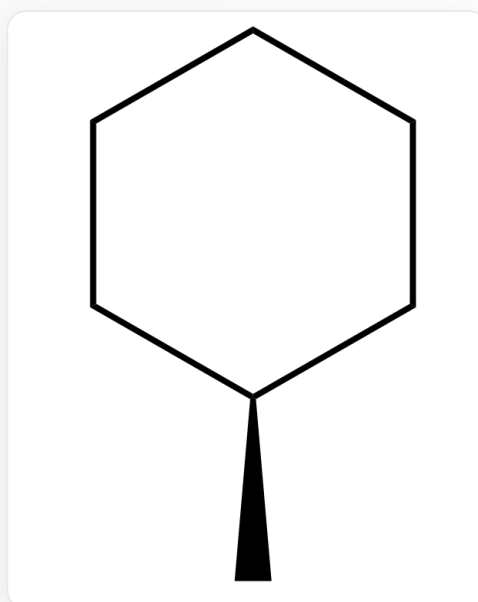
Has enantiomers



C[C@H]1C[C@H](C)CC1

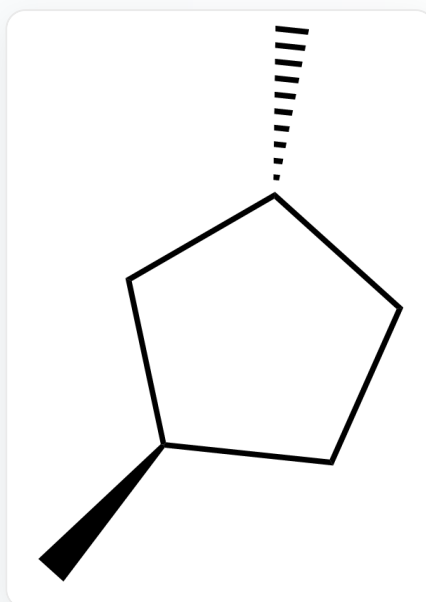
Has enantiomers

G.



CC1CCCCC1

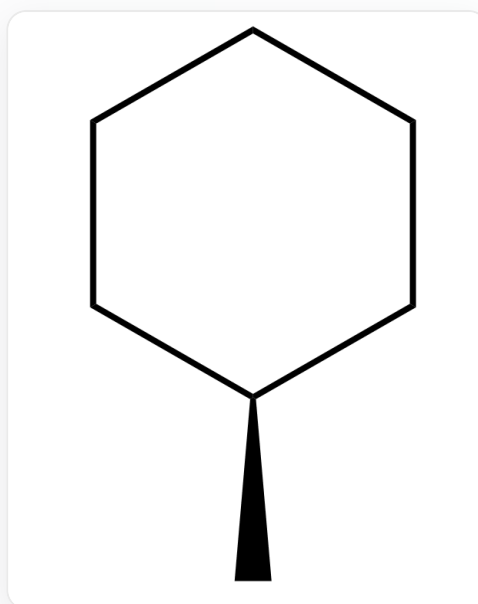
No enantiomers



C[C@H]1C[C@H](C)CC1

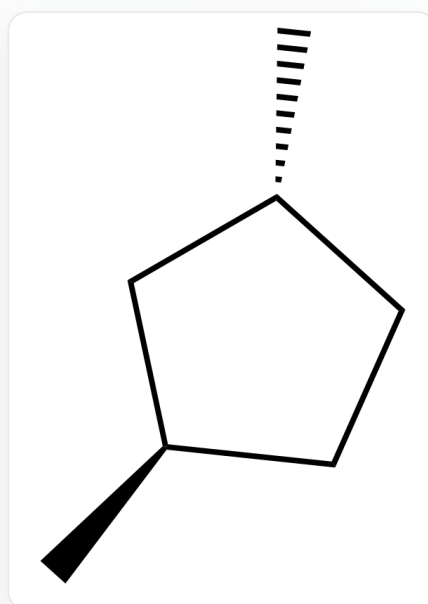
No enantiomers

H.



CC1CCCCC1

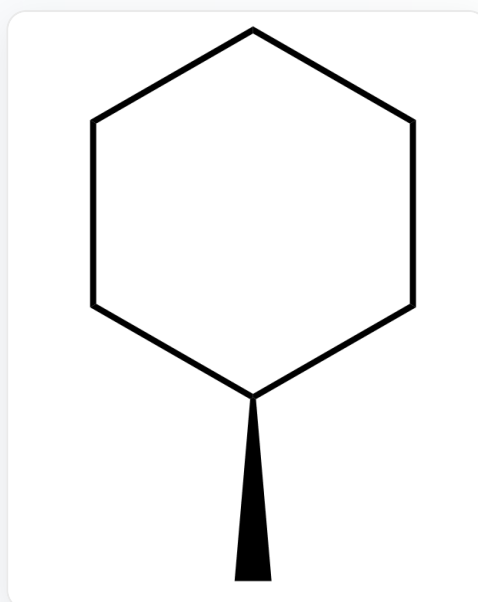
No enantiomers



C[C@H]1C[C@H](C)CC1

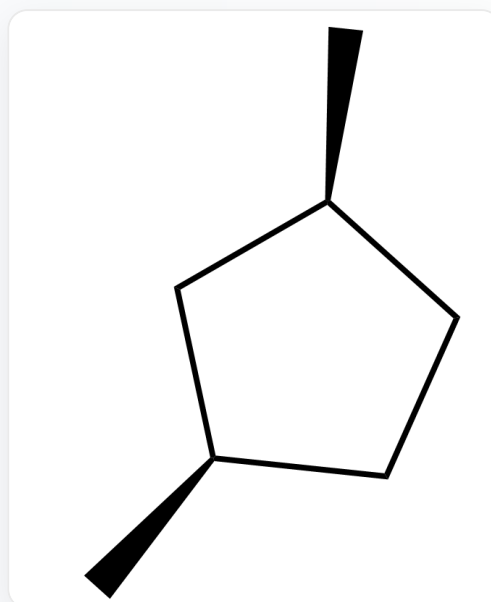
Has enantiomers

I.



CC1CCCCC1

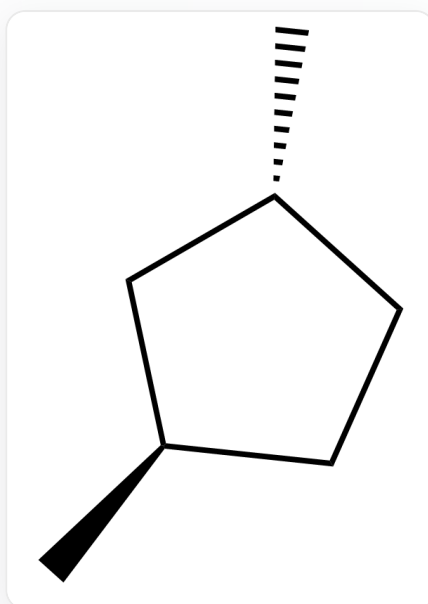
No enantiomers



C[C@@H]1C[C@H](C)CC1

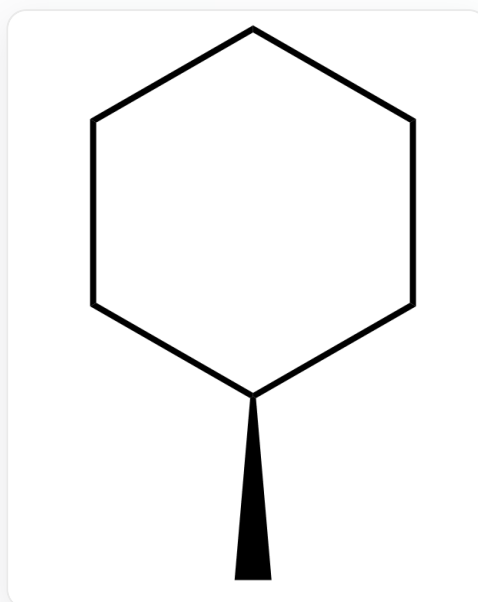
No enantiomers

J.



C[C@H]1C[C@H](C)CC1

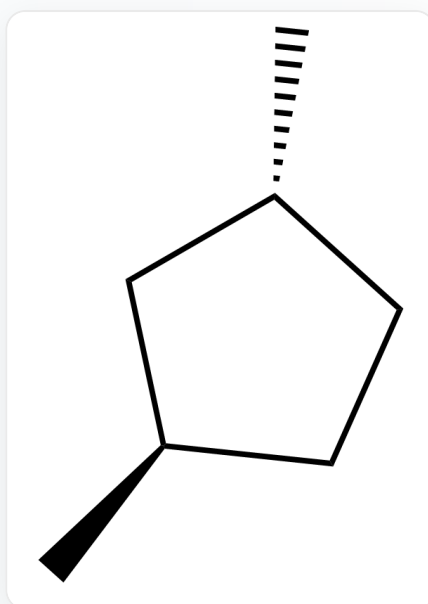
Has enantiomers



CC1CCCCC1

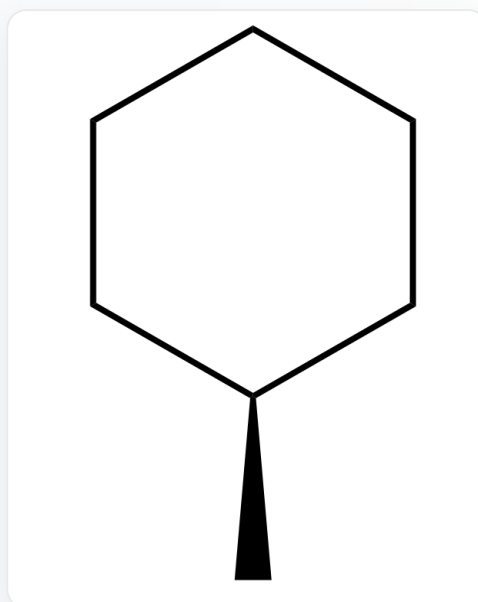
No enantiomers

K.



C[C@H]1C[C@H](C)CC1

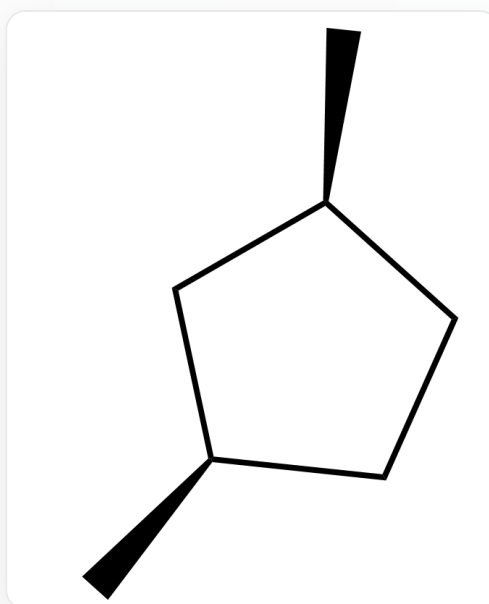
No enantiomers



CC1CCCCC1

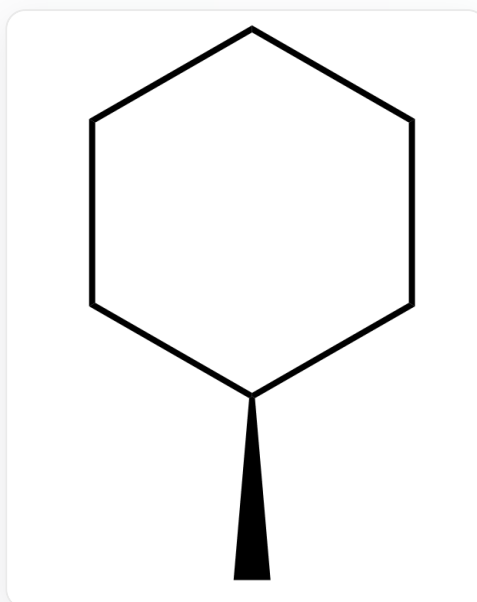
No enantiomers

L.



C[C@@H]1C[C@H](C)CC1

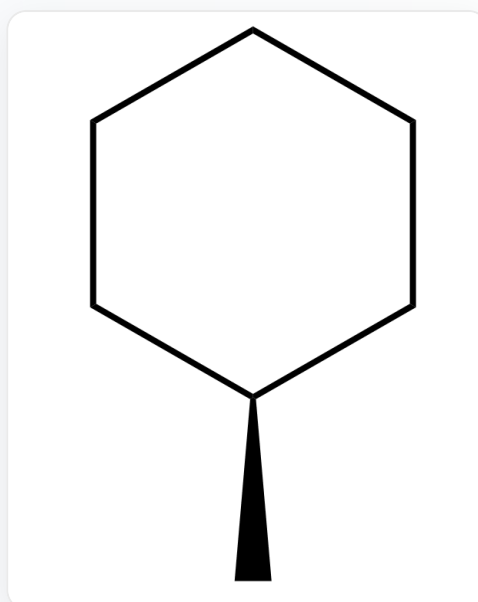
No enantiomers



CC1CCCCC1

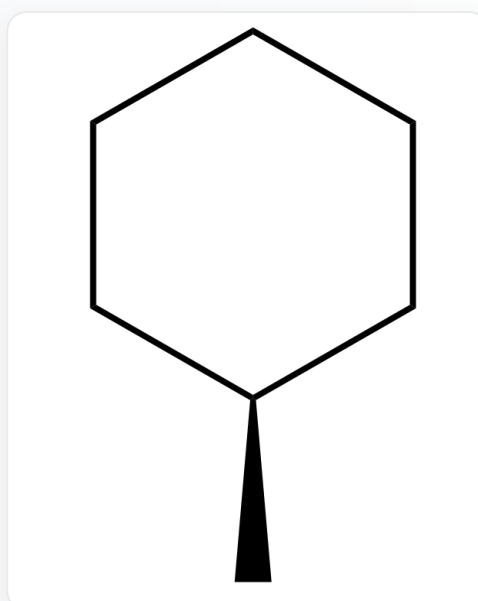
No enantiomers

M.



CC1CCCCC1

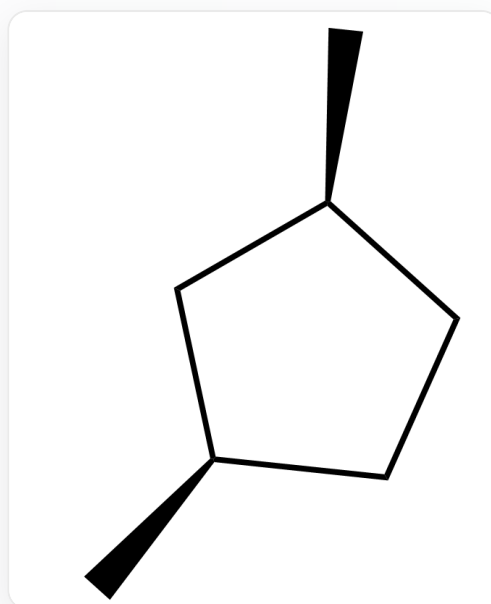
No enantiomers



CC1CCCCC1

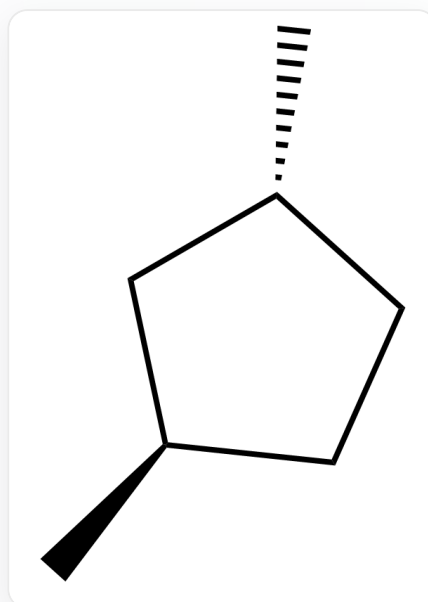
No enantiomers

N.



C[C@@H]1C[C@H](C)CC1

No enantiomer



C[C@H]1C[C@H](C)CC1

No enantiomer

Answer

Correct Answer: **A**

Detailed Explanation

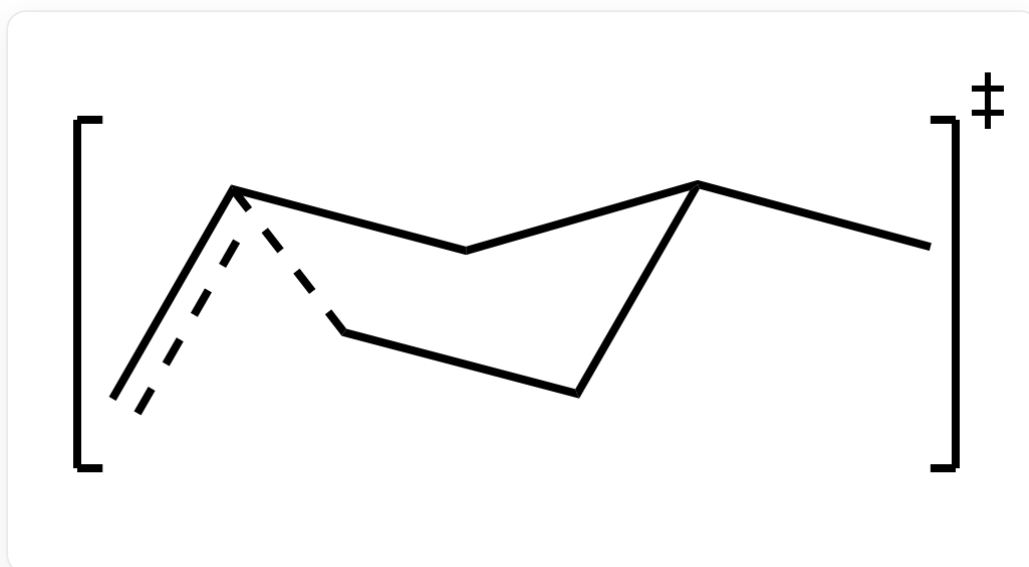
The cyclization of alkyl radicals is less reversible and tends to yield the kinetically favored five-membered ring product.

CHECKPOINT

2 PTS

The cyclization of alkyl radicals is less reversible and tends to yield the kinetically favored five-membered ring product

For the cyclization of the first substrate, the transition state energy is lower when the methyl group is in the equatorial position.



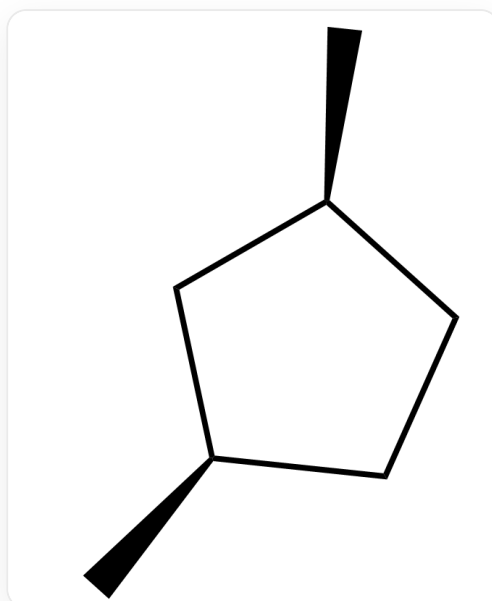
cc1C[C@H](C)CC1

CHECKPOINT

1 PTS

For the cyclization of the first substrate, the transition state energy is lower when the methyl group is in the equatorial position

Therefore, the first reaction yields the cis product.


C[C@@H]1C[C@H](C)CC1

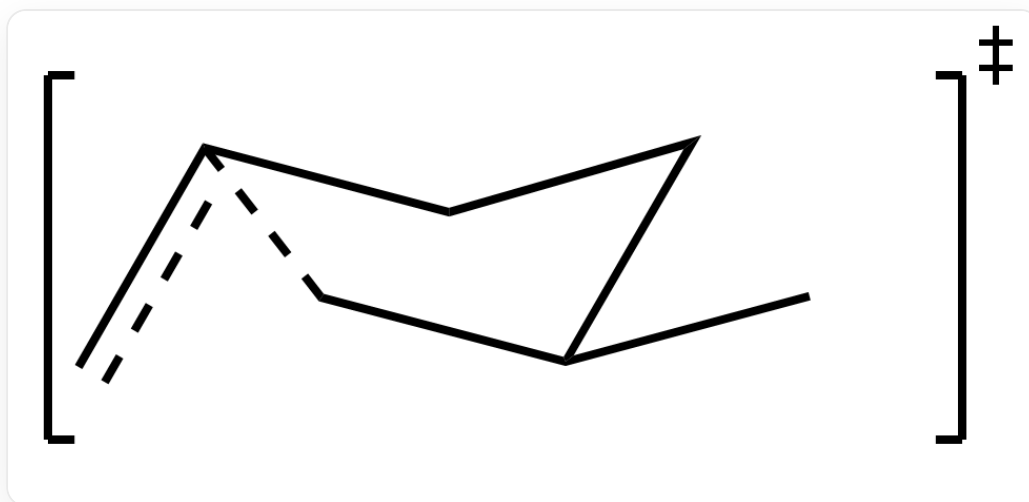
This product contains a mirror plane and thus does not exhibit enantiomerism.

CHECKPOINT

1 PTS

This product contains a mirror plane and thus does not exhibit enantiomerism

For the cyclization of the second substrate, the transition state energy is also lower when the methyl group is in the equatorial position.



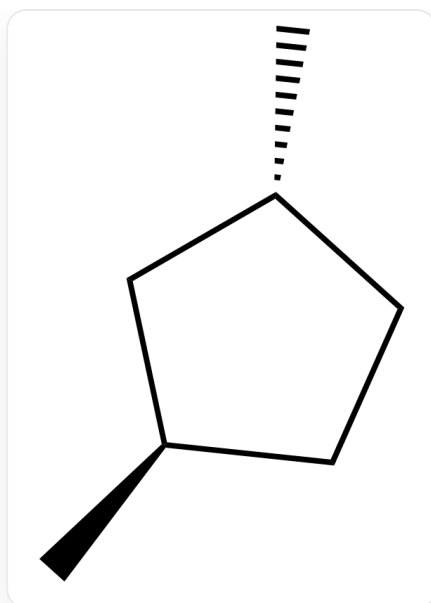
cc1CC[C@@H](C)C1

CHECKPOINT

1 PTS

For the cyclization of the second substrate, the transition state energy is also lower when the methyl group is in the equatorial position

Therefore, the second reaction yields the trans product.



C[C@H]1C[C@H](C)CC1

This product lacks a mirror plane, inversion center, or improper rotation axis and is thus optically active.

CHECKPOINT

1 PTS

This product lacks a mirror plane, inversion center, or improper rotation axis and is thus optically active

In summary, the correct answer to this question is A.